

Statement Purpose:

Discussion about ordinary differential equations Euler method.

Activity Outcomes:

Students will understand and implement ordinary differential equations in matlab

Instructor Note:

solve the exercise given at the end of the notes

1- Euler Method (First Order Taylor Series Method)

- An equation that consists of derivatives is called a differential equation.
- First order Taylor series method is known as Euler Method. Euler's method is a **numerical technique** to solve ordinary differential equations of the form

$$\frac{dy}{dx} = f(x, y), y(0) = y_0$$

Euler Method

$$y_{i+1} = y_i + h f(x_i, y_i)$$

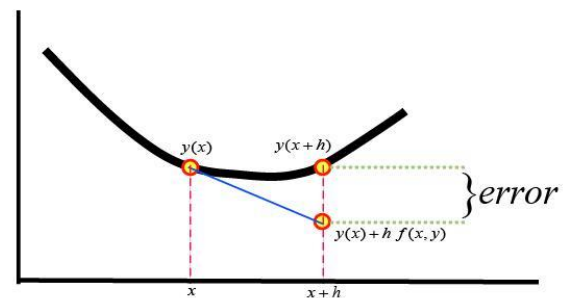
Algorithm

Input: Continuous function $f(x, t)$
Initial value $x_0 = x(t_0)$,
Final time
 t_f
Step h

Output: A function $x(t)$ satisfies $dx(t)/dt = f(x, t)$

Steps:

1. $x_0 = x(t_0)$
2. Loop
 $x_i = x_{i-1} + h f(x_{i-1}, t_{i-1})$
End



loop

Example

Solve $\frac{dx(t)}{dt} + 2x(t) = 1$ for $x_0 = x(0) = 1, h = 0.01$ and $tf = 0.1$.

Analytical Solution

`>> x = dsolve('Dx = 1- 2 *x')`

$x = C2 / (2 * \exp(2*t)) + 1/2$

we know $x(0)=1 \rightarrow C2 = 1$ and $x(t) = 0.5 (1 + \exp(-2t))$

Numerical Solution

iteration	t	x	$f(x_{i-1}, t_{i-1}) = -2x+1$	Analytical Solution $x(t)=0.5(1+\exp(-2t))$	Difference x- x_analytically
0	0	1	-1	1	0
1	0.01	0.9900	-0.9800	0.9901	-0.0993e-003
2	0.02	0.9802	-0.9604	0.9804	-0.1947e-003
3	0.03	0.9706	-0.9412	0.9709	-0.2863e-003
4	0.04	0.9612	-0.9224	0.9616	-0.3741e-003
5	0.05	0.9520	-0.9039	0.9524	-0.4583e-003
6	0.06	0.9429	-0.8858	0.9435	-0.5390e-003
7	0.07	0.9341	-0.8681	0.9347	-0.6164e-003
8	0.08	0.9254	-0.8508	0.9261	-0.6904e-003
9	0.09	0.9169	-0.8337	0.9176	-0.7612e-003
10	0.10	0.9085		0.9094	-0.8290e-003



Program

```
function z=euler(f,t0,tf,h,x0)

% Solve Ordinary Differential Equations using Euler
Method
% f is function
% t0 and x0 are initial values
% tf is final time
% h is step

k=1;
y(k)=x0;
ty(k) = t0;
for t= t0+h : h :tf
k=k+1;
y(k)=y(k-1)+h*feval(f, y(k-1));
ty(k) = t;
end
z = [ty' y'];
```

Call the program

```
>> fx=@(x)1-2*x
>> z=euler(fx,0,0.1,0.01,1)
```

z =

0	1.0000
0.0100	0.9900
0.0200	0.9802
0.0300	0.9706
0.0400	0.9612
0.0500	0.9520
0.0600	0.9429
0.0700	0.9341
0.0800	0.9254
0.0900	0.9169
0.1000	0.9085

